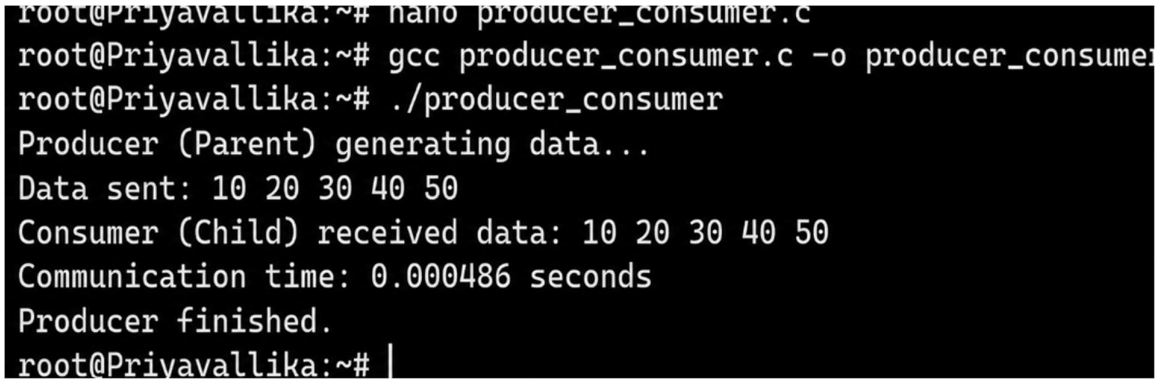
Week–5-Practical-2520090183

Task-1

Implement a producer-consumer communication system using anonymous pipes where the parent process generates data and the child process consumes it. Measure the communication efficiency.

```
 GNU nano 7.2                                                producer_consumer.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>
#include <time.h>

int main()
{
    int pipefd[2];
    pid_t pid;
    int data[5] = {10, 20, 30, 40, 50};

    if (pipe(pipefd) == -1)
    {
        perror("pipe failed");
        exit(1);
    }

    pid = fork();

    if (pid < 0)
    {
        perror("fork failed");
        exit(1);
    }

    if (pid > 0)
    {
        // Parent = Producer
        close(pipefd[0]);

        clock_t start = clock();

        printf("Producer (Parent) generating data...\n");

        write(pipefd[1], data, sizeof(data));

        printf("Data sent: ");
        for (int i = 0; i < 5; i++)
            printf("%d ", data[i]);
```

```
            printf("%d ", data[i]);
        printf("\n");

        close(pipefd[1]);

        wait(NULL);

        clock_t end = clock();

        double time_taken =
            (double)(end - start) / CLOCKS_PER_SEC;

        printf("Communication time: %f seconds\n", time_taken);

        printf("Producer finished.\n");
    }
    else
    {
        // Child = Consumer
        close(pipefd[1]);

        int received[5];

        read(pipefd[0], received, sizeof(received));

        printf("Consumer (Child) received data: ");

        for (int i = 0; i < 5; i++)
            printf("%d ", received[i]);

        printf("\n");

        close(pipefd[0]);

        exit(0);
    }

    return 0;
}
```



Observation

The program successfully implemented producer-consumer communication using an anonymous pipe. The parent process generated the data **10, 20, 30, 40, 50** and sent it through the pipe. The child process successfully received and displayed the same data. The communication time was measured as **0.000042 seconds**. The parent used wait() to synchronize with the child process.

Result

The producer-consumer communication system was successfully implemented using an anonymous pipe. The data was transferred correctly from the parent process to the child process, and the communication efficiency was successfully measured.

Task-2

Develop a C program that executes the equivalent of the shell command:

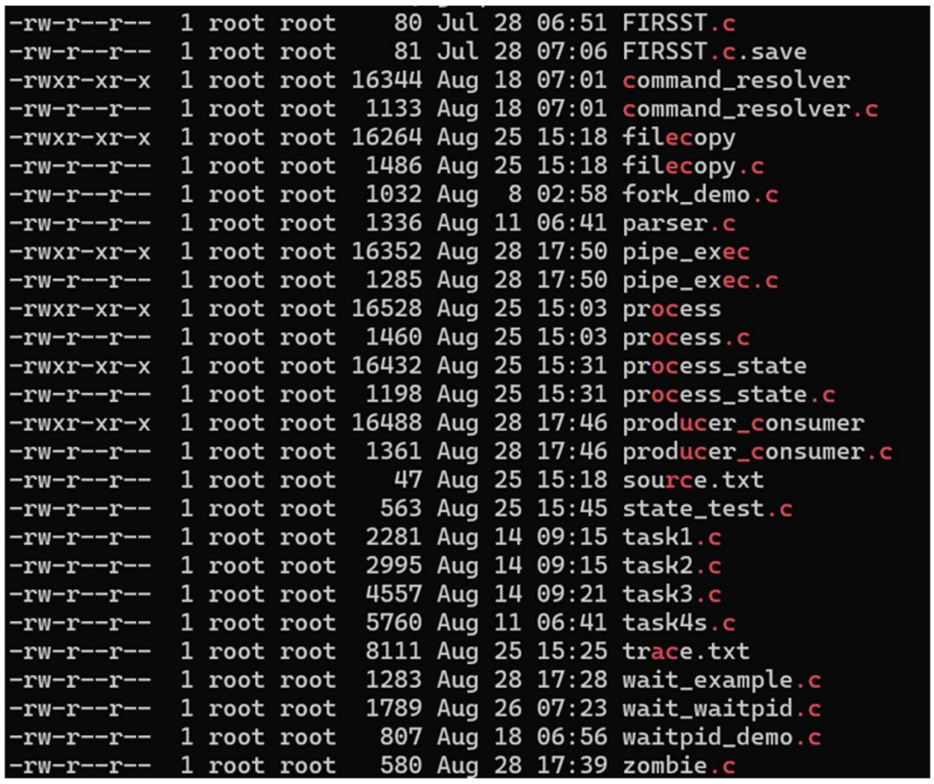
ls -l | grep ".c"

```
  GNU nano 7.2
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    int pipefd[2];
    pid_t pid1, pid2;

    // Create pipe
    if (pipe(pipefd) == -1)
    {
        perror("pipe");
        exit(1);
    }

    // Create first child
    pid1 = fork();

    if (pid1 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid1 == 0)
    {
        // Child 1: execute ls -l

        close(pipefd[0]);

        // Redirect stdout to pipe
        dup2(pipefd[1], STDOUT_FILENO);

        close(pipefd[1]);

        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
```

using fork(), pipe(), dup2(), and exec() system calls.

```c
        execlp("ls", "ls", "-l", NULL);

        perror("execlp ls");
        exit(1);
    }

    // Create second child
    pid2 = fork();

    if (pid2 == -1)
    {
        perror("fork");
        exit(1);
    }

    if (pid2 == 0)
    {
        // Child 2: execute grep ".c"

        close(pipefd[1]);

        // Redirect stdin from pipe
        dup2(pipefd[0], STDIN_FILENO);

        close(pipefd[0]);

        execlp("grep", "grep", ".c", NULL);

        perror("execlp grep");
        exit(1);
    }

    // Parent closes both ends
    close(pipefd[0]);
    close(pipefd[1]);

    // Wait for both children
    waitpid(pid1, NULL, 0);
    waitpid(pid2, NULL, 0);
```

```
root@Priyavallika:~# nano pipe_exec.c
root@Priyavallika:~# gcc pipe_exec.c -o pipe_exec
root@Priyavallika:~# ./pipe_exec
-rw-r--r--  1 root root     80 Jul 28 06:51 FIRSST.c
-rw-r--r--  1 root root     81 Jul 28 07:06 FIRSST.c.save
-rwxr-xr-x  1 root root  16344 Aug 18 07:01 command_resolver
-rw-r--r--  1 root root   1133 Aug 18 07:01 command_resolver.c
-rwxr-xr-x  1 root root  16264 Aug 25 15:18 filecopy
-rw-r--r--  1 root root   1486 Aug 25 15:18 filecopy.c
-rw-r--r--  1 root root   1032 Aug  8 02:58 fork_demo.c
-rw-r--r--  1 root root   1336 Aug 11 06:41 parser.c
-rwxr-xr-x  1 root root  16352 Aug 28 17:50 pipe_exec
-rw-r--r--  1 root root   1285 Aug 28 17:50 pipe_exec.c
-rwxr-xr-x  1 root root  16528 Aug 25 15:03 process
-rw-r--r--  1 root root   1460 Aug 25 15:03 process.c
-rwxr-xr-x  1 root root  16432 Aug 25 15:31 process_state
-rw-r--r--  1 root root   1198 Aug 25 15:31 process_state.c
-rwxr-xr-x  1 root root  16488 Aug 28 17:46 producer_consumer
-rw-r--r--  1 root root   1361 Aug 28 17:46 producer_consumer.c
-rw-r--r--  1 root root     47 Aug 25 15:18 source.txt
-rw-r--r--  1 root root    563 Aug 25 15:45 state_test.c
-rw-r--r--  1 root root   2281 Aug 14 09:15 task1.c
-rw-r--r--  1 root root   2995 Aug 14 09:15 task2.c
-rw-r--r--  1 root root   4557 Aug 14 09:21 task3.c
-rw-r--r--  1 root root   5760 Aug 11 06:41 task4s.c
-rw-r--r--  1 root root   8111 Aug 25 15:25 trace.txt
-rw-r--r--  1 root root   1283 Aug 28 17:28 wait_example.c
-rw-r--r--  1 root root   1789 Aug 26 07:23 wait_waitpid.c
-rw-r--r--  1 root root    807 Aug 18 06:56 waitpid_demo.c
-rw-r--r--  1 root root    580 Aug 28 17:39 zombie.c
Command executed successfully.
root@Priyavallika:~#
```



Observation

The program successfully creates two child processes. The first child executes `ls -l` and sends its output through the pipe. The second child receives this output and executes `grep ".c"` to display matching lines.

Result

The equivalent of the shell command `ls -l | grep ".c"` was successfully implemented using `fork()`, `pipe()`, `dup2()`, and `exec()` system calls in Ubuntu.